

Cloud computing infrastructure

[00:00](#) Hello, cloud enthusiast!

[00:02](#) Get ready to learn exactly what cloud computing is all about, including how it works, the components of a cloud infrastructure, and different cloud service models.

[00:12](#) So, first up, what exactly is cloud computing?

[00:17](#) Cloud computing is the practice of using on-demand computing resources as services hosted over the internet.

[00:24](#) “Over the internet” is what makes up the cloud part.

[00:28](#) It eliminates the need for organizations to find, set up, or manage resources themselves.

[00:34](#) And they only pay for what they use.

[00:36](#) Cloud computing uses a network to connect users to a cloud platform.

[00:41](#) This is a virtual space where they can access and borrow computing services.

[00:45](#) A primary computer handles all communication between devices and servers to share information.

[00:51](#) And there are privacy and security measures to keep everything safe.

[00:55](#) Here's another way to think about it.

[00:58](#) Picture cloud computing like a shared kitchen in a rental apartment, owned by a property management company or, in the case of the cloud, a third-party service host.

[01:08](#) The kitchen has many appliances and cooking tools, just as the cloud platform has servers, storage, hardware, and software.

[01:16](#) So, when someone in the apartment gets hungry, they just go ahead and cook a meal in the well-equipped kitchen.

[01:21](#) They don't each have to buy their own wooden spoons or toaster oven.

[01:25](#) Likewise, the cloud enables organizations to access computing resources on-demand, without spending time and money buying and maintaining their own storage, hardware, and software.

[01:36](#) It's the unique infrastructure of a cloud computing model that makes all of this possible.

[01:42](#) This infrastructure has four main components: hardware, storage, network, and virtualization.

[01:49](#) Let's check out hardware first.

[01:52](#) Types of hardware include servers, processors, and memory; network switches, routers, and cables; firewalls and load balancers; cooling systems; and power supplies.

[02:06](#) These are the physical items needed to keep things running.

[02:09](#) Now, data storage in a cloud computing infrastructure can occur in three main ways: file, object, or block.

[02:18](#) File storage keeps data in one place and organizes it in a simple, easy-to-understand way through a hierarchy of files in folders.

[02:26](#) This is the oldest and most widely used data storage system, but it's a bit cumbersome and can only accomplish so much.

[02:34](#) Next, object storage holds unstructured data, along with its metadata.

[02:40](#) Metadata is just data about data.

[02:42](#) For example, a picture taken with a smartphone might contain information about the location, the date, and the type of device that captured the image.

[02:52](#) This is really useful for understanding the photo, just as metadata explains what its own data is all about.

[02:58](#) Lastly, block storage divides large volumes of data into smaller pieces, optimally organized, with unique labels.

[03:06](#) An advantage of block storage is that the data is easily accessible, but it can be expensive and has limited capability to handle metadata.

[03:15](#) All right, now we have the network.

[03:18](#) After all, cloud computing infrastructure needs a way to connect to its backend resources, and this connection is made possible through a network of the physical hardware.

[03:28](#) Through this network, users tap into cloud resources using some of the hardware mentioned earlier, including routers and firewalls.

[03:37](#) Basically, the physical network setup is what enables the virtual one to operate.

[03:43](#) Finally, there is virtualization, which is technology that creates a virtual version of physical infrastructure, like servers, storage, and networks.

[03:53](#) This is what lets the service work without a connection.